

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q20-1458

Speed 138 kV  
March 01, 2022

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1458. The Interconnection Customer proposes a 89.6 MW solar photovoltaic facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on April 16, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 403 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,659,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: GAGA / BARNHAM 138 KV line  
Contingency: SILVER BAY POWER / SPEED 138 KV  
Loading: 125.6% (Rating: 190 MVA)  
DFAX: 100.0%  
Pre-project loading: 78.5%
2. Facility: SPEED / GAGA 138 KV  
Contingency: SILVER BAY POWER / SPEED 138 KV L  
Loading: 123.8% (Rating: 209 MVA)  
DFAX: 100.0%  
Pre-project loading: 80.8%
3. Facility: SILVER BAY POWER / SPEED 138 KV  
Contingency: SPEED / GAGA 138 KV  
Loading: 121.9% (Rating: 212 MVA)  
DFAX: 100.0%  
Pre-project loading: 79.6%
4. Facility: BARNHAM / NAROD 138 KV line  
Contingency: SILVER BAY POWER / SPEED 138 KV  
Loading: 114.4% (Rating: 191 MVA)  
DFAX: 100.0%  
Pre-project loading: 67.5%

#### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

#### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

#### ***4. Required Network Upgrades and Cost Estimates***

| Facility                        | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------|---------------------------------------------|---------------|-----------------------|
| BARNHAM / GAGA 138 KV           | Rebuild line with higher-capacity conductor | \$2,556,000   | 32% (\$825,000)       |
| SPEED / GAGA 138 KV             | Rebuild line with higher-capacity conductor | \$3,788,000   | 32% (\$1,223,000)     |
| SILVER BAY POWER / SPEED 138 KV | Rebuild line with higher-capacity conductor | \$21,669,000  | 32% (\$6,996,000)     |
| BARNHAM / NAROD 138 KV line     | Rebuild line with higher-capacity conductor | \$20,487,000  | 32% (\$6,615,000)     |
|                                 | Total allocated to project                  |               | \$15,659,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.